



# **Data Analytics in Motorsport: Predicting Formula 1 Race Outcomes**

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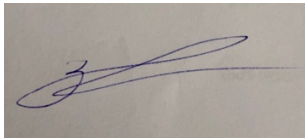
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# Declaration

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A handwritten signature in blue ink, appearing to be 'Giorgi Chachava', on a light-colored background.

**Giorgi Chachava**

Word length: 10,020 words

# 1 Executive Summary

This dissertation explores how data analytics and machine learning can be used to understand and predict race outcomes in Formula 1. The main goal of this research was to build models that can predict whether a driver will finish inside the Top 10 and estimate their exact finishing position. These predictions are important not only from a sporting perspective but also from business and strategic point of view, because race results influence sponsorship value, team earnings, development priorities and global exposure.

To achieve this, I used the “Formula 1 World Championship (1950-2024)” dataset from Kaggle, which contains detailed information about drivers, constructors, race results, qualifying sessions, sprint races, pit stops and circuits. The dataset was merged and cleaned to create a single race-level table where each row represents one driver in one race. Additional features, such as driver experience and recent form were engineered to capture performance patterns that do not appear directly in the raw data.

Before building models, I conducted exploratory data analysis to understand the relationships within the dataset. The EDA showed that qualifying position is one of the strongest predictors of race outcomes, confirming long-known patterns in the sport. Pit stop strategy also had clear effect with drivers making more stops generally finishing lower in the race. Long-term team performance showed predictable dominance cycles, with teams like Ferrari, Mercedes and Red Bull leading different eras. Driver performance analysis revealed that modern scoring systems and longer seasons strongly influence total career points.

For the modelling stage, I first created baseline models (Logistic Regression for Top 10 prediction and Linear Regression for finishing position). These models provided a reference point to evaluate improvement. The baseline classification model reached an accuracy of around 0.74, while the baseline regression model achieved an R2 value of around 0.33.

More advanced models were then tested, including Random Forest and Gradient Boosting. Gradient Boosting performed the best in both tasks. With engineered features included, the Top 10 classifier achieved an accuracy of about 0.83 and the regression model reached an R2 of around 0.63. SHAP analysis showed that recent form, qualifying position, pit stop duration and experience were among the most influential variables in predicting race outcomes.

The business insights drawn from the modelling highlights several practical patterns. Consistent qualifying performance remains one of the biggest competitive advantages. Recent driver form plays a stronger role than expected, aligning with psychological and momentum-based patterns observed in real Formula 1. Driver experience contributes to greater stability, while pit stop efficiency still influences race outcomes in meaningful ways.

These insights can help teams with strategic decision-making and provide sponsors with a clearer understanding of where competitive value comes from.

In conclusion, this dissertation demonstrates that race outcomes in Formula 1 can be predicted with a reasonable degree of accuracy using historical data and machine learning techniques. The results provide both technical and business value, showing that performance in the sport reflects a combination of engineering strength, human factors and strategic execution. Future improvements could include adding weather data, tire-level information or lap-by-lap telemetry to further enhance prediction accuracy.

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## 2 Introduction

Formula 1 (F1) is the highest level of motorsport – a sport where technology, data and human performance come together. Over the years, F1 has changed from being just about fast cars and skilled drivers to becoming a world of data and analytics. Each race weekend produces massive amounts of information such as lap times, tire data, pit stops, weather conditions and sensor readings from cars. According to Mitchell (2022), a Formula 1 team can generate up to 11.8 billion data points in one season. Teams use these data points to improve car design, plan strategies and make real-time race decisions. This shows how important data analysis has become in modern motorsport.

Predicting the results of an F1 race is not just a technical task – it also has a strong business side. Technically, it involves using data science and machine learning to understand how different factors such as driver skill, team strength or track characteristics affect results. From a business point of view, race outcomes shape how much teams earn, the size of their sponsorships and how much attention they get from fans and media. As Patil et al. (2022) explain, data analytics gives motorsport teams a competitive advantage because it turns raw data into meaningful insights that can improve both performance and business decisions.

In Formula 1, only top 10 finishers in a race earn points for the championship (FIA, 2025). The winner gets 25 points, the second-place driver 18 and the tenth-place driver earns one point. This point system decides who wins the Drivers' and Constructors' Championships and directly affects the money and reputation of the teams (Porzio, 2025). Teams that regularly finish in the points get more prize money and attract better sponsors. While this dissertation will not only focus on predicting top 10 finishes, understanding the points system helps show why predicting race results is valuable from a business and strategic point of view.

The main goal of this dissertation is to predict how drivers will finish in upcoming races using historical data and to understand which factors most affect performance. These factors include driver skill, team performance, qualifying positions and circuit characteristics. To achieve this, I will use historical data and apply different machine learning methods such as Random Forest and Logistic and Linear Regression. The data will be cleaned and prepared before analysis and I will use Python libraries such as pandas, scikit-learn and SHAP. The models will be evaluated using metrics such as accuracy, F1-score and Root Mean Square Error (RMSE). I will also use feature importance and SHAP values to interpret results and understand which factors most strongly influence performance. This combination of prediction and explanation will give both technical and business insights.

The dataset for this project will be the “Formula 1 World Championship (1950 – 2024)” dataset from Kaggle (Kaggle, 2025). It includes details about drivers, teams (constructors),

races, qualifying results, pit stops and circuits. With over 70 years of history, it provides a rich source of information for finding patterns and building practical models. The data will be explored carefully to find trends – for example, how much qualifying position affects the final result or how team performance changes over time. A clear data preparation process, exploratory analysis, well-tested machine learning models will help ensure the findings are reliable.

From a business perspective, Formula 1 is a multi-billion-dollar global industry. It generates huge revenue through sponsorships, broadcasting rights and advertising. According to Brittle C. (2025), F1 generated around \$3.65 billion in revenue in 2024, making its strongest financial performance yet. Because of this, accurate predictions and performance insights are valuable not only to teams but also to sponsors, investors and media partners. For example, a team could use predictions to evaluate drivers before signing them or sponsors could decide which team offers the best marketing value. Analytics can also help improve fan engagement – many fans today enjoy seeing data-driven predictions and performance breakdowns that make races more interactive.

Beyond the racetrack, Formula 1 occupies a significant place in the global economy and the sports-entertainment business model. Each Grand Prix weekend attracts thousands of international visitors, stimulates hospitality and tourism industries, and offers host regions substantial marketing and infrastructure opportunities. For example, a study found that the 2021 Dutch Grand Prix generated an additional 44.5 million euros in spending in the Amsterdam metropolitan region and directly supported nearly 500 local jobs (Formula 1, 2022). Moreover, race weekends create a valuable platform for host cities and sponsors to enhance brand visibility, leverage dynamic pricing for accommodation and hospitality and generate long-term economic ripple effects via the “multiplier effect.” Scholars also argue that the synergy of technology, media and sport inherent in Formula 1 amplifies its influence well beyond the track, contributing to innovation, job creation, foreign direct investment and destination branding (Ross, n.d.).

The new 2026 Formula 1 regulations highlight how the sport continues to evolve. Barretto (2024) says that FIA announced changes that will make cars lighter, faster, and more efficient. These include a new hybrid engine system and aerodynamic updates designed to make racing closer and more sustainable. Such changes will make data analysis even more important, as teams will have to adapt to new technical conditions. Predictive analytics can help teams understand how these new factors affect performance and how to adjust their strategies accordingly. This shows how data-driven decision-making is becoming central not only to race engineering but also to business planning in F1.



This dissertation will combine technical and business perspectives. On the technical side, it will apply machine learning to analyze complex data and predict race outcomes. On the business side, it will connect these predictions to practical uses such as sponsorship decisions, marketing and strategic planning. The study aims to show that analytics in motorsport can create real business value. By understanding what drives success on the track, teams and stakeholders can make smarter decisions off the track.

In conclusion, this research will explore how data analytics can be used to predict Formula 1 race results and understand key performance factors. It will provide both technical findings and business insights for one of the most competitive and technologically advanced sports in the world. Predicting race outcomes is not only about accuracy – it is also about understanding the dynamics of success in a sport where performance, innovation and business are deeply connected.

## 3 Literature Review

Formula 1 has always been a sport defined by engineering, competition and speed, but in the last decade it has also become deeply connected to data science, machine learning, psychology and global economics. Because this dissertation aims to predict race outcomes while also explaining their business relevance, the literature review brings together several research areas: performance analytics, machine learning studies, sporting rules, human performance, operational strategy and economic impact. Each domain highlights a different dimension of what influences results in Formula 1 and why prediction models matter in both sporting and commercial terms.

### 3.1 Data, Technology and Performance Analysis in Formula 1

Modern Formula 1 operates at the intersection of engineering and real-time data analysis. The sport relies heavily on telemetry, simulation and sensor-driven insights to understand performance at every stage of a race weekend. Mitchell (2022) describes how an F1 car can produce around 1.5 terabytes of data over a single race and teams generate more than 11 billion data points across a season. This scale of information is necessary because race performance depends on a complex mix of aerodynamics, tire behavior, mechanical balance, engine modes and environmental conditions.

Mitchell (2022) also explains how the digital „pipeline“ inside a team works. Data is collected through hundreds of sensors, combined with computational fluid dynamics (CFD)

simulations and wind tunnel measurements and processed using automation tools like Alteryx to track costs, analyze upgrades and correlate predictions with track behavior. This is important because Formula 1 teams operate under a cost cap, meaning efficient decision-making is now part of competitive performance. The literature repeatedly emphasises that F1 is not simply about designing fast cars – it is about interpreting huge quantities of information correctly and making strategic decisions supported by quantitative evidence.

Another important point raised by Mitchell (2022) is that F1 teams no longer rely solely on track testing. Because regulations limit wind tunnel time and computational resources, teams must extract insights from data correlations rather than large-scale experimentation. This creates a natural link with machine learning: algorithms can help teams detect non-obvious patterns between variables such as tire degradation, fuel load effects, aerodynamic upgrades or weather conditions (Whitehead, 2025).

Overall, the existing literature clearly shows that the modern era of Formula 1 is inseparable from data-driven performance analysis. For this dissertation, this context supports the idea that predictive modelling is not only possible, but also aligned with how F1 teams already work internally.

### 3.2 Machine Learning and Predictive Analytics in Motorsport

Academic research on Formula 1 performance prediction is still limited, but several studies show the potential of machine learning for analysing complex sporting competitions. One of the most relevant works in this area is the study by Patil et al. (2022), which performs a systematic, data-driven analysis of Formula 1 race features using correlation, PCA (Principal Component Analysis) and linear regression. Their research uses race characteristics from 2015-2019 – such as tire usage, pit stops, position-based metrics and driver classification rates – to identify which factors influence total championship points.

Patil et al. (2022) find that variables like average pole position, tire strategy and completed laps correlate strongly with performance, while PCA reveals that most meaningful race information can be explained through a few principal components related to driver position, tire behavior and penalties. Importantly, they also highlight that there is almost no published research analysing race outcomes using predictive modelling. Most previous sports analytics studies focus on football, basketball or horse racing, which means Formula 1 remains under-explored academically.

This gap supports the purpose of this dissertation. While Patil et al. use regression to explain final points over a full season, their work does not attempt to predict race-level finishing

positions for individual drivers, nor does it incorporate contemporary variables such as sprint races, real-time pit metrics or psychological performance trends. The absence of models predicting driver-level race outcomes stands out as a clear research opportunity.

Another notable observation from Patil et al. (2022) is that tire type and qualifying position significantly influence total points. These findings align with common knowledge in Formula 1 but also validate that the models used in this dissertation should place strong weight on grid position and race execution variables.

In summary, existing literature demonstrates that machine learning can extract meaningful insights from Formula 1 data, but the field lacks studies that predict race outcomes at the individual driver level. This dissertation directly addresses this gap.

### 3.3 Race Format, Qualifying and Sporting Regulations

Race outcomes in Formula 1 are shaped by the sporting format, regulations and aerodynamic principles that determine how cars behave on track. Several sources provide detailed explanations of these systems and highlight why certain variables matter more than others in performance prediction.

#### 3.3.1 Qualifying and Grid Position

Clark (2024) explains how the qualifying format – Q1, Q2 and Q3 – creates a knockout system that identifies the fastest cars and determines their starting positions. Because overtaking is limited on many circuits, qualifying performance becomes one of the strongest predictors of race results. Clark notes that even small differences in grid position matter, especially on circuits with tight layouts where overtaking requires large tire deltas or strategic alternatives. This source is valuable because it reinforces why grid and qualifying positions are included as key predictive variables in the modelling stages.

#### 3.3.2 Sprint Races

Kristensen (2022) describes the introduction of sprint races, which change the weekend structure by adding a shorter race used to set the grid for the Grand Prix. Sprint races also award points and influence driver momentum. Importantly for this dissertation, sprint results represent a measurable performance signal because they show how drivers and cars

behave under race conditions without pit stops. Including sprint-related data reflects a modernised understanding of performance.

### 3.3.3 Dirty Air and Aerodynamic Behavior

Cook (2025) provides a clear explanation of „dirty air“, the disturbed airflow behind an F1 car that reduces downforce and makes overtaking more difficult. With the leading teams closer in pace in 2025, Cook argues that dirty air has re-emerged as a major limiting factor in overtaking opportunities. This means drivers starting behind a closely matched car struggle to pass without significant tire or strategy differences. For predictive modelling, this reinforces why grid position consistently acts as a dominant variable.

### 3.3.4 Technical Regulations

Several major changes in F1 regulations also influence performance models. The FIA (2014) leaflet explains the shift to hybrid V6 power units and how energy recovery systems (ERS-K and ERS-H) changed the structure of engine performance. Meanwhile, Barretto (2024) highlights the upcoming 2026 regulations, which include lighter cars, increased battery power and active aerodynamics. Evidence from Formula 1 history suggests that regulatory changes shape competitive dynamics over time, influencing team dominance cycles and the relative importance of different performance variables (Straw, 2022). This context is important for interpreting long-term patterns in the dataset.

Across these sources, the literature consistently highlights one message: race results are strongly influenced by technical and procedural systems such as qualifying, sprint formats, aerodynamic constraints and regulation cycles. These factors justify the inclusion of qualifying position, grid position and year-based variables in predictive models.

## 3.4 Human Factors, Driver Psychology and Performance Consistency

While Formula 1 is an engineering-driven sport, driver psychology and human performance play a significant role in race outcomes. Hynes (2024) provides a detailed look at mental preparation, driver confidence and the psychological cycles that influence race-to-race performance. This is particularly relevant for predictive modelling because patterns such as „recent form“ or „short-term performance momentum“ often originate from psychological rather than technical variables.

Drivers like Daniel Ricciardo describe how confidence loss can trigger negative spirals, while rebuilding mental stability leads to improved results. Sergio Perez similarly uses a sports psychologist to recover form after poor performances. Hynes (2024) also explains that elite drivers often rely on routines such as visualisation, deep focus sessions or structured warm-ups to enter what they call a „flow state“ (Brolin, 2020). These insights clarify why performance in Formula 1 tends to cluster across several races rather than fluctuate randomly.

The literature also shows how driver experience affects consistency. Experienced drivers tend to manage pressure better, adapt more effectively and provide more stable results across long seasons. Performance coaches quoted by Hynes (2024) confirm that familiarity with race environments and preparation routines can reduce variability in outcomes.

This context supports including engineered features such as driver experience and recent form in the predictive models. These variables capture psychological or behavioral patterns that are not directly present in the dataset but strongly influence race results. Moreover, the literature suggests that such factors can shape both short-term momentum and long-term competitive trajectories.

### 3.5 Pit Stop Operations and Strategic Performance

Pit stops are a critical operational aspect of Formula 1, with direct influence on race outcomes. Barretto (2019) provides an inside view of pit stop execution, highlighting that top teams perform over 1,000 practice stops per season. Every member of the pit crew has a specific task and the entire system is optimised to achieve both speed and consistency. Barretto explains that the difference between a perfect stop and a poorly executed one can be less than a second, yet that single second can drastically affect finishing positions.

Several important insights from Barretto (2019) are relevant for this dissertation:

- Consistency often matters more than absolute speed. A uniformly quick set of pit stops avoids unnecessary time loss.
- Track-specific characteristics (such as pit lane layout or surface inclination) influence execution.
- Pit stop failures or delays directly correlate with worse race outcomes.
- Technological investment (wheel guns, jacks, control systems) plays a significant role in maintaining elite performance.

Because pit stops affect race position volatility, the literature supports including variables such as pit stop count and average pit stop duration in prediction models. Although pit stops are not the strongest predictors compared to qualifying, they still meaningfully influence the middle phase of races where strategic decisions unfold.

### 3.6 Economic and Business Dimensions of Modern Formula 1

Formula 1 is not only a sport – it is also a global commercial ecosystem. The financial literature highlights how economic factors shape team performance, competitive dynamics and the relevance of predictive analysis.

Brittle (2025) reports that F1 revenue rose to approximately \$3.65 billion in 2024, driven by media rights, sponsorship deals and promotional contracts. Growth in fan attendance, social media engagement and F1 TV subscriptions reinforces the sport's commercial trajectory. As the financial value of the sport grows, accurate performance insights become more important for stakeholders such as sponsors, broadcasters and investors.

From a host city perspective, Grand Prix weekends significantly boost local economies. The study published by Formula 1 (2022) on the Dutch Grand Prix shows 44.5 million Euros in additional regional spending and hundreds of jobs created. Academic research also supports these findings, showing that hosting Formula 1 events leads to a measurable increase in tourism demand in host countries (Ramasamy and Yeung, 2020). Ross (n.d.) further explains how F1 races contribute to tourism, infrastructure investment and business visibility. These findings show that race results, driver popularity and competitive narratives directly influence economic outcomes for cities, teams and advertisers.

This economic context justifies the business relevance of race outcome prediction:

- Sponsors want to know which teams or drivers offer the best exposure.
- Teams use predictive models for driver evaluation, car development or strategic planning.
- Media platforms benefit from analytics-driven fan engagement.

The literature therefore supports the idea that predictive analytics in Formula 1 has genuine commercial value beyond sports performance.

### 3.7 Summary of Gaps and Positioning of This Dissertation

The existing literature shows that Formula 1 performance is influenced by technological systems, sporting rules, human factors and economic pressures. However, several notable gaps remain:

1. Lack of driver-level race prediction models – Patil et al. (2022) analyse features statistically but do not predict finishing positions for individual drivers.
2. Limited integration of modern variables – Sprint races, pit stop timing, psychological momentum and updated regulations are rarely included in previous studies.
3. Absence of machine learning models that combine technical, strategic and human factors – No current study includes engineered features such as driver experience or recent form.
4. Minimal research connecting prediction accuracy with business value.

This dissertation fills these gaps by:

- Using machine learning algorithms to predict both Top 10 finishes and exact race positions.
- Incorporating modern features reflecting current F1 dynamics.
- Applying SHAP values to explain the importance of different factors.
- Linking predictive findings to business insights relevant for teams and sponsors.

Together, the literature supports the methodological choices of the dissertation and highlights the need for a modernised, data-driven and business-aware approach to understanding Formula 1 race outcomes.

## 4 Methodology – Data Collection and Preprocessing

The dataset used for this project is the “Formula 1 World Championship Dataset (1950-2024)”, published on Kaggle (Kaggle, 2025). It contains detailed historical data about Formula 1 races, drivers, constructors, circuits, qualifying results, pit stops, sprint races and other related tables. The dataset provides information at different levels – from race-level summaries to lap-by-lap timings – which makes it a valuable source for both exploratory and predictive analysis. For this research, the data was downloaded in CSV format and processed in Python using libraries such as pandas, NumPy and scikit-learn in Google Colab.

The goal of the preprocessing stage was to build a single, clean dataset where each row represents the performance of a driver in a specific race. To achieve this, I identified the key tables that form the analytical foundation and the supporting tables that add additional predictive features. The main or “key” tables are results, races, drivers, constructors and circuits. These were chosen because they contain the essential attributes needed to link and describe each race outcome. The results table is the core dataset, as it contains the finishing positions, points, laps, and status for each driver-race combination. The races table adds contextual information such as the season year, round number and circuit, while drivers and constructors provide the descriptive and categorical information about participants and teams. The circuits table includes geographical information (country, location, latitude and longitude), which can later be used for regional or environmental insights if needed.

The merging process followed a step-by-step approach. I first joined the results table with races using the shared key `raceld`, and then added drivers, constructors and circuits successively through their respective identifiers. Each join used a left merge to make sure no valid race records were dropped. During this process, I carefully inspected the merged dataset for duplicates, missing values and inconsistencies. Some races had multiple rows for the same driver due to historical data variations or overlapping entries (for example, reclassifications or shared drivers in older seasons). To ensure one valid record per driver-race, I applied a simple rule-based deduplication: when multiple entries existed for the same (`raceld`, `driverId`) pair, I kept the one with the highest points, and if there was a tie, the one with the lowest finishing position and most laps completed. This logic guarantees consistency while maintaining the most meaningful record for analysis.

After constructing the main table, I added three supporting datasets to create new features that could potentially improve prediction accuracy. From the qualifying table, I extracted the driver’s qualifying position, which is a strong indicator of race performance as it directly determines the starting grid. From `sprint_results`, available since 2021 (Kristensen, 2022), I added sprint position and sprint points to reflect driver performance in sprint sessions, which can influence both confidence and grid placement. Finally, from the `pit_stops` table, I aggregated two features: the total number of pit stops and the average pit stop duration (converted from milliseconds to seconds for interpretability). These features capture strategic and operational aspects of racing, such as pit efficiency and tire management, which are known to affect race outcomes.

Not all tables in the dataset were used. Several were excluded intentionally to avoid redundancy or data leakage. The `driver_standings` and `constructor_standings` tables, for example, contain post-race summaries that are derived from results, meaning they already include outcome information. Using them would artificially inflate model accuracy because



they indirectly reveal what we are trying to predict. Similarly, the `constructor_results` table duplicates team-level performance data already represented through the constructor IDs in the results table. Other tables, such as `seasons` and `status`, were excluded because they provide metadata (like URLs or text descriptions) that are not relevant to modeling race outcomes. The `lap_times` table, although rich in detail, was not used in this phase because it provides lap-level granularity, resulting in millions of records. Since the aim of this project is to predict final race results rather than lap-by-lap behavior, using it would unnecessarily complicate the analysis at this stage.

After merging and feature creation, I checked the final dataset for missing values and column consistency. The structure of the resulting table ensures that each record uniquely represents one driver's performance in one race, identified by a combination of `raceld` and `driverId`. Non-essential or redundant columns, such as URLs or repeated session timestamps (e.g., free practice dates), were removed to reduce noise. Finally, the clean dataset was saved, which serves as the foundation for predictive modeling and evaluation phases presented in the next sections.

## 5 Exploratory Data Analysis (EDA)

The purpose of the exploratory data analysis stage was to examine main patterns and relationships within the Formula 1 dataset before applying machine learning models. This step helps to understand how different features such as qualifying positions, pit stop strategies and team or driver performance contribute to final race results. Visualization plays an important role in this phase, because it makes complex relationships easier to interpret and allows for initial business and sporting insights to emerge. The analysis was performed using Python with the `pandas`, `seaborn` and `matplotlib` libraries and focused on key aspects relevant to both racing strategy and team competitiveness.

### 5.1 Dataset Overview

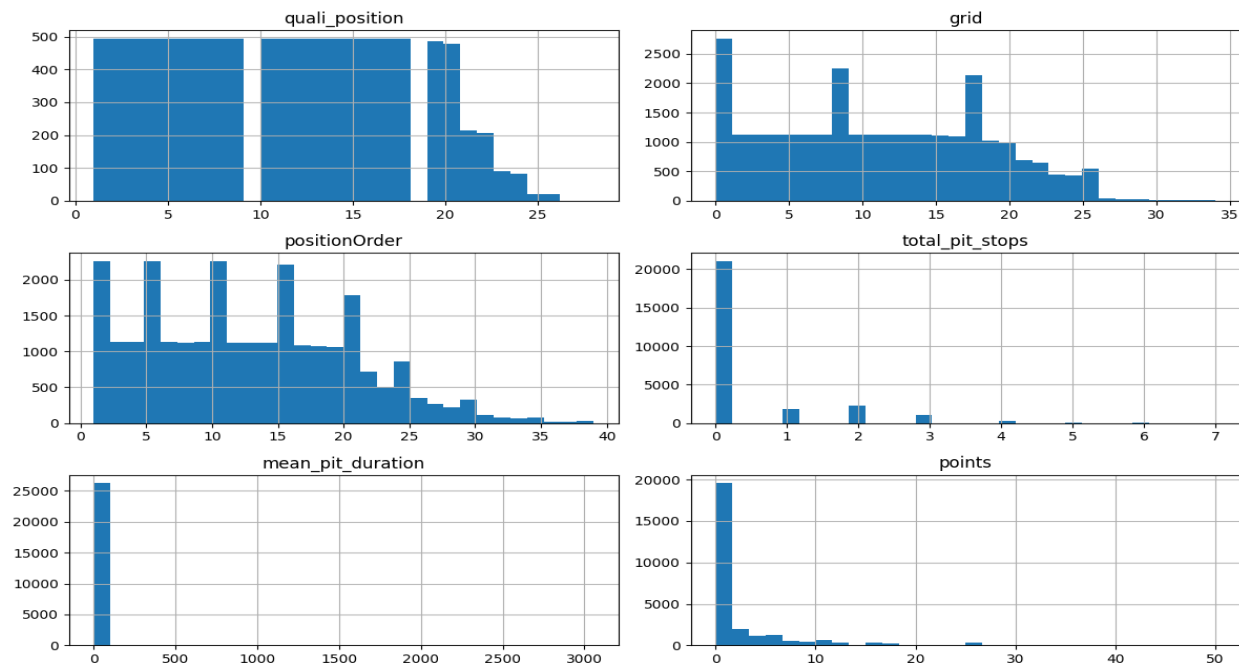
After combining all key and supporting tables, the final dataset contained over 26,000 rows and 45+ features, representing one driver in one race. Most variables were numeric, while a few were dates or categorical fields like driver names and constructor references.

Basic descriptive statistics helped confirm that the dataset behaves as expected. Grid and qualifying positions ranged across a full starting order, finishing positions (`positionOrder`) showed the usual right-skewed pattern where only small number of drivers regularly finish near the front and most finish in midfield or lower positions. Points were heavily skewed

toward zero, which reflects the fact that only the top ten finishers score under modern rules. Pit stop-related features also showed realistic values: most races involved one or two stops, and average pit stop duration sat within typical modern F1 ranges.

### 5.1.1 Feature Distributions

To visualize this more clearly, histograms were plotted for several key features: `quali_position`, `grid`, `positionOrder`, `total_pit_stops`, `mean_pit_duration` and `points`. These plots helped identify how the dataset behaves in general. For example, qualifying and grid positions were well spread across the field, finishing positions showed the expected concentration in lower positions and pit stop values highlighted that strategic variation is quite limited across most races. Points again showed a sharp drop-off after the top finishers.



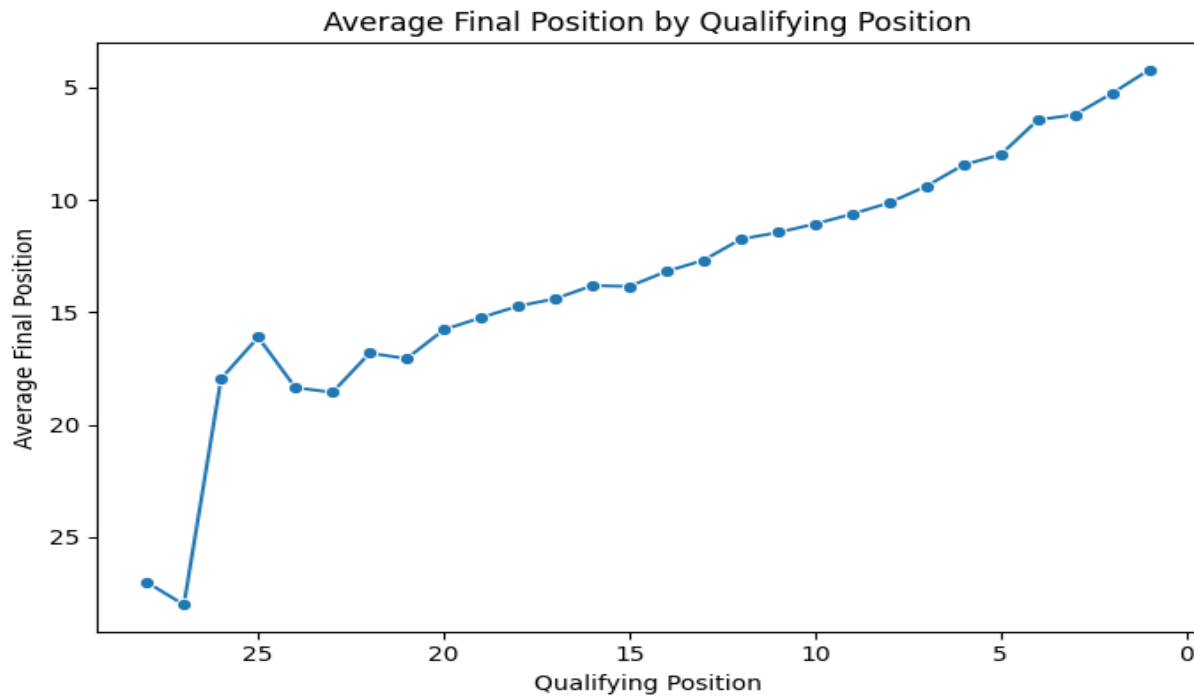
**Figure 1. Feature Distribution Histograms**

### 5.1.2 Outlier Checks

Potential outliers were also checked using Z-scores. A few outliers appeared in pit-stop-related variables and in finishing positions, which makes sense in motorsport because unusual race scenarios (retirements, accidents, long pit delays) naturally create extreme values. Since these are real racing situations – not errors – they were kept in the dataset.

## 5.2 Qualifying Position and Race Outcome

Qualifying performance has long been considered one of the strongest predictors of race success in Formula 1. To analyze this relationship, the average finishing position of drivers was plotted against their qualifying position.

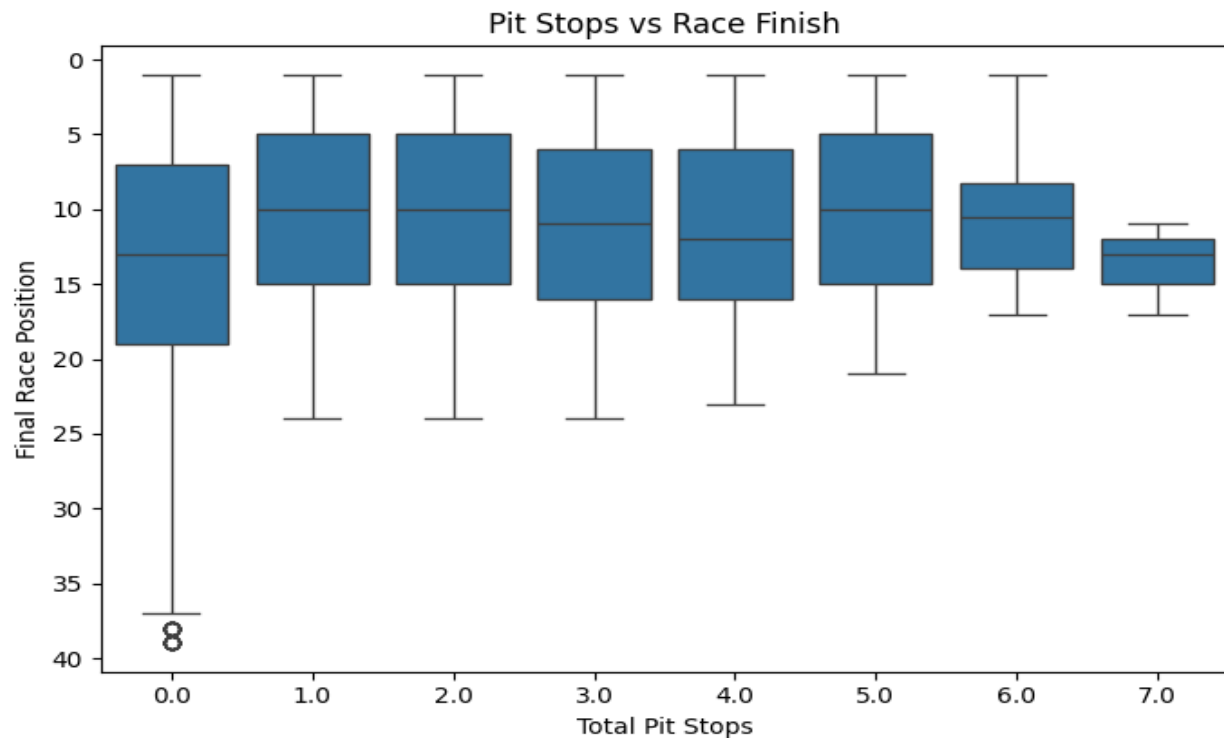


**Figure 2. Relationship between Qualifying and Final Race Position**

The results show a clear downward trend, indicating that drivers starting from the front of the grid are far more likely to finish in stronger positions. This visual confirms the competitive advantage of starting closer to pole position. The relationship can be explained by the reduced likelihood of being caught in traffic or incidents during the race start. As Patil et al. (2022) note, starting position is one of the key factors with a statistically significant impact on race outcomes, which aligns with the patterns observed in this analysis. From a business perspective, qualifying performance not only predicts results but also affects sponsorship visibility and brand exposure, as front-running cars receive more broadcast time and attention.

### 5.3 Pit Stop Strategy and Race Performance

Pit stop efficiency and strategy are critical operational factors in Formula 1. To explore this, a boxplot was created to show how the total number of pit stops correlates with final race positions.

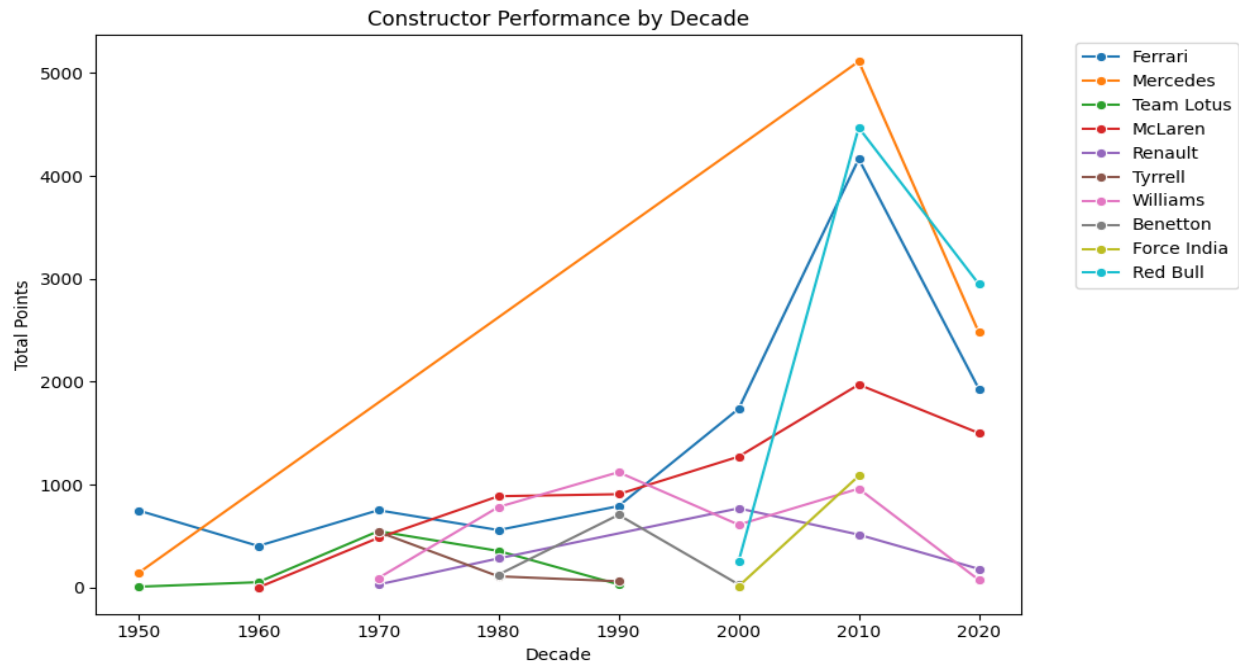


**Figure 3. Effect of Pit Stops on Final Race Position**

The boxplot reveals that most winning or top-finishing drivers complete one or two pit stops, while those with three or more generally finish lower in the order. The median finishing position gradually worsens as pit stops increase, suggesting that additional stops often result from tire degradation, damage or suboptimal strategies. However, there are a few outliers – cases, where multiple pit stops still led to competitive results – indicating successful aggressive strategies or safety car advantages. These findings highlight the balance between tire management and race pace, reinforcing that operational decisions significantly affect performance outcomes and, indirectly, the financial results of teams.

## 5.4 Constructor Dominance Across Decades

To understand long-term patterns in performance, total constructor points were aggregated by decade. This analysis captures how team dominance has evolved alongside regulatory and technological changes.

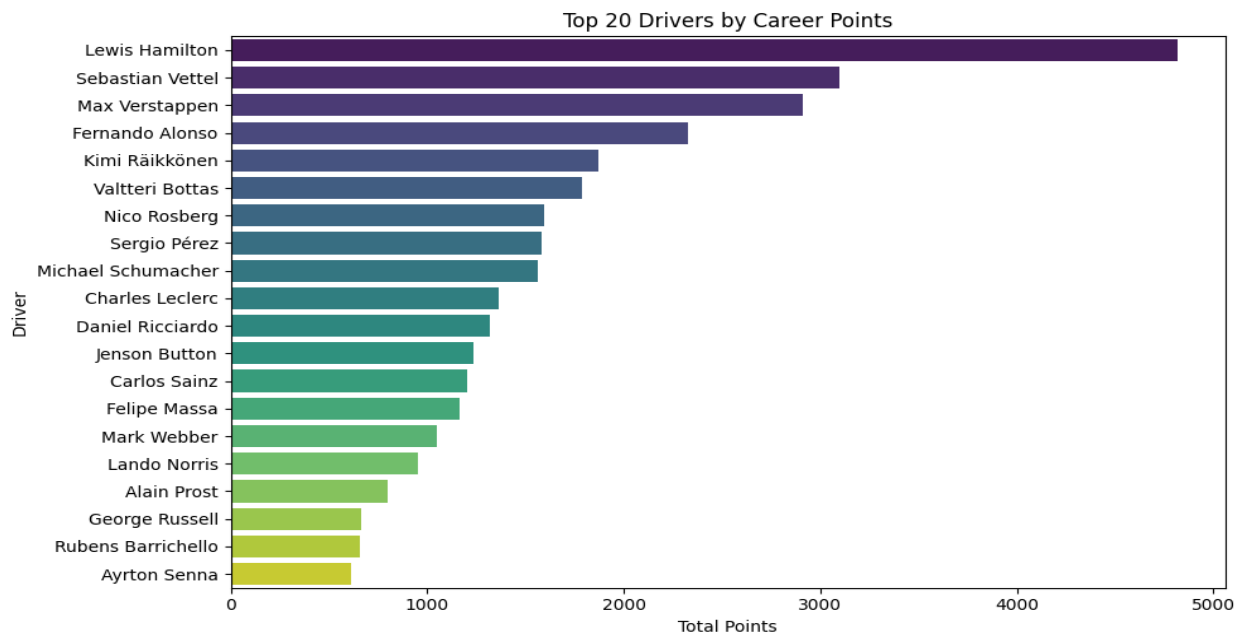


**Figure 4. Constructor Performance by Decade**

The visual shows distinct eras of dominance. Ferrari demonstrates consistent competitiveness throughout most decades, while Mercedes achieved remarkable success during the 2010s under the hybrid power regulations introduced in 2014 (FIA, 2014). Red Bull emerges as the leading constructor of the 2020s, reflecting the strength of their chassis and driver pairing. In earlier decades, dominance was more evenly distributed, suggesting that competition was less concentrated among a few teams. These observations align with historical trends documented by FIA (2025) and highlight the cyclical nature of Formula 1 competitiveness. From a business perspective, periods of team dominance correlate with higher sponsorship revenue and global brand recognition, as seen with Ferrari's long-standing commercial appeal and Mercedes' surge in visibility during their dominant years.

## 5.5 Driver Career Performance Distribution

Individual driver performance over time was examined by aggregating total career points for each driver. The top 20 drivers were visualized using a bar chart to illustrate differences in long-term success.

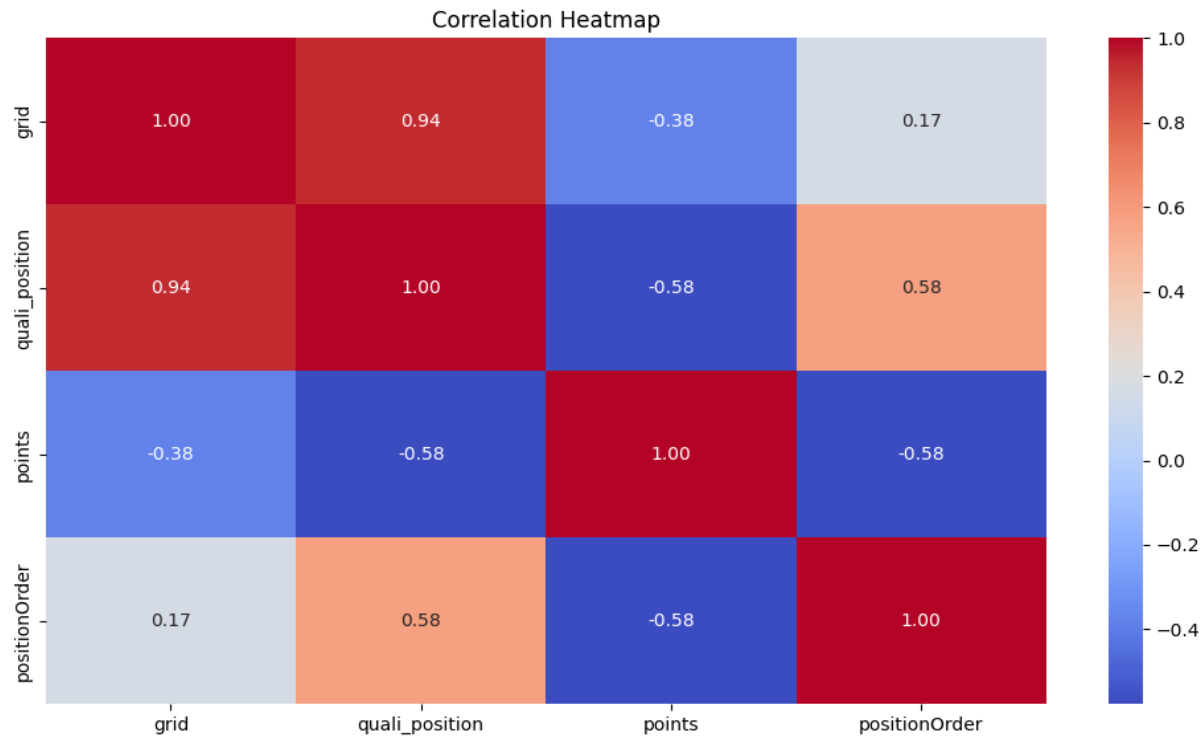


**Figure 5. Top 20 Drivers by Career Points**

The chart reveals a strong skew toward modern drivers such as Lewis Hamilton and Sebastian Vettel, who led by a large margin. This pattern can partly be explained by the introduction of longer seasons and higher scoring systems in modern Formula 1. As Brittle (2025) reports, Formula 1's commercial success has grown alongside an expanded calendar of 24 races. More races per season naturally give modern drivers more opportunities to score points compared to earlier eras. While this may limit direct comparisons with earlier eras, it still highlights the consistent excellence required to remain competitive across multiple seasons. This also reflects the growing importance of driver value in attracting sponsors and media coverage, making driver success not only a sporting but also financial asset.

## 5.6 Correlation Matrix of Key Variables

Finally, a correlation heatmap was used to examine the relationships between core quantitative variables: grid, qualifying position, race points and final race order.



**Figure 6. Correlation Matrix of Key Variables**

The heatmap demonstrates strong positive correlations between qualifying and finishing positions (0.58), confirming that starting near the front typically leads to better results. Points show a negative correlation with qualifying and grid positions. This suggests that initial positioning remains the most influential factor in determining results. These findings are consistent with the broader understanding of Formula 1 race dynamics, where qualifying speed and early track position often outweigh strategic adjustments made during the race (Mitchell, 2022).

## 5.7 Summary of Insights

Overall, the exploratory data analysis confirms that Formula 1 results are shaped by a mix of technical performance, operational strategy and long-term investment. Qualifying success is the strongest single predictor of finishing position, while efficient pit stop strategies and

consistent team performance over time also play critical roles. The findings reflect how sporting outcomes are tightly linked to business performance in Formula 1, influencing sponsorship returns, brand exposure and fan engagement. The next section of the dissertation will build on these insights by developing predictive models to estimate future race outcomes and identify the most influential performance factors.

## 6 Methodology - Algorithms & Evaluation Plan

After preparing and exploring the dataset, the next step of the project is to build models that can predict Formula 1 race outcomes. Since the goal of the dissertation is two-sided – predicting whether a driver will finish in the Top 10 and predicting the exact finishing position – the modelling process is also divided into two parts: a classification task and a regression task. In this section, I explain both why these tasks are used, why specific algorithms were chosen as baselines and how the models are evaluated.

### 6.1 Why two tasks: Classification and Regression

Formula 1 race results can be interpreted in different ways. The first approach is to predict whether a driver finishes in the Top 10. This is important because only the top ten finishers score points and points are directly connected to championship standings, team earnings and sponsorship value. Predicting Top 10 finishes gives a clear business-related insight: it separates drivers who score from those who don't, which is useful for teams, analysts and sponsors.

The second approach is to predict the exact finishing position of each driver. This task is more detailed and helps understand not only “who scores points” but also “how competitive each driver actually is”. Exact finishing position predictions can help teams evaluate performance over time and understand the effect of qualifying, strategy and race conditions on results.

Because both views are useful and represent different levels of detail, this dissertation uses both a classification model and a regression model.



## 6.2 Baseline Algorithms: Logistic Regression and Linear Regression

Before testing more advanced models, it is important to build simple and interpretable baseline models. These baselines help set a reference point, so we can later compare how much improvement we get from models like Random Forest.

For the classification task, I use Logistic Regression. It is a standard algorithm for binary classification problems and is widely used, because it is simple, fast and easy to interpret (GeeksforGeeks, 2025a). Logistic Regression works well when baseline relationships between features and the target are straightforward. Since the goal here is to predict whether a driver finishes in the Top 10, Logistic Regression is a logical starting point.

For the regression task, I use Linear Regression as the baseline model. The algorithm predicts a continuous value, which in this case is the finishing position. Linear Regression is also simple and easy to understand (GeeksforGeeks, 2025a). Although race outcomes are influenced by many nonlinear factors, it is helpful to see how a basic linear model performs before using more advanced machine learning techniques.

These baseline models do not aim to be perfect. Instead, they help show how much of the prediction problem can be solved using basic methods and how much improvement is needed.

## 6.3 Advanced Algorithms: Random Forest and Gradient Boosting

After baseline models are defined, the next step is to test more advanced machine learning algorithms and examine whether they can capture more complex patterns in the data. Formula 1 race outcomes depend on many factors, which are not always well represented by simple linear relationships.

The first advanced model is Random Forest, which builds multiple decision trees and combines their predictions, which allows it to model non-linear relationships and interactions between variables. It's widely used for tabular datasets and is generally robust to overfitting, making it a natural next step after baseline models (Staff, 2025b).

After Random Forest, Gradient Boosting is applied. Unlike Random Forest, Gradient Boosting builds trees sequentially, where each new model focuses on correcting the errors made by previous ones. This approach is well suited for problems where many features contribute small but important effects, which is often the case in Formula 1 performance data. Because of this, Gradient Boosting is expected to provide stronger predictive performance than both baseline and Random Forest models (GeeksforGeeks, 2025b).

To further improve model performance, limited hyperparameter tuning will be applied to the Gradient Boosting models using RandomizedSearchCV (scikit-learn, n.d.c). This approach tests a predefined number of random parameter combinations and selects the configuration that performs best on the training data. Randomized search was chosen instead of exhaustive grid search because it provides a good balance between performance improvement and computational efficiency. The final Gradient Boosting models will use the best-performing parameter set identified during this tuning process.

## 6.4 Train-Test Split and Avoiding Data Leakage

To evaluate the models properly, the data is divided into training and test sets. The training data is used to fit the model, and the test data is used to measure how well the model performs on unseen races. An 80/20 split is used, meaning 80% of the data goes to training and 20% to testing (scikit-learn, n.d.a).

For the classification task, I used stratified sampling, which keeps the proportion of Top 10 and non-Top 10 drivers the same in both sets. This makes the evaluation more reliable, especially since classes are not perfectly balanced.

It is important to avoid data leakage – when the model accidentally learns information from the test set during training. To prevent this, all preprocessing steps such as scaling are fitted only on the training data.

## 6.5 Feature Scaling with StandardScaler

Some algorithms, especially Logistic Regression, work better when all numeric features are on a similar scale. For example, qualifying position might be around 1-20, while year values go from 1950 to 2024. These differences can make training unstable.

For this reason, I use StandardScaler, which transforms each feature to have a mean of zero and a standard deviation of one. The scaler is always fitted only on the training data and then applied to both training and testing sets. This makes the models more stable and helps the optimization process run smoothly (scikit-learn, n.d.b).

## 6.6 Evaluation Metrics for Classification

For the classification model (Top 10 prediction), I use four main metrics:

- Accuracy – how often the model predicts correctly.
- Precision – out of all the drivers predicted as Top 10, how many actually finished in the Top 10.
- Recall – out of the real Top 10 finishers, how many the model successfully identified.
- F1-score – A balance between precision and recall (Friedman, 2024).

These metrics give a full picture of how the model behaves. Accuracy alone is not enough because the dataset is slightly imbalanced, so F1-score is especially important. I also include a confusion matrix to see where the model makes mistakes (Staff, 2025a) – for example, how many Top 10 drivers it misses or how many incorrect Top 10 predictions it makes.

## 6.7 Evaluation Metrics for Regression

For the regression model (predicting finishing position), I use three common metrics:

- RMSE (Root Mean Squared Error) – Measures how far the predictions are from real values, giving more weight to large errors.
- MAE (Mean Absolute Error) – Shows the average absolute error between predicted and actual positions.
- R2 (Coefficient of Determination) – Shows how much of the variation in finishing positions the model can explain (Chugh, 2020).

These metrics work well because race results are continuous values and errors of several positions are expected. RMSE is useful for showing how large the typical mistake is, while MAE gives simpler average error. R2 helps understand whether the model is capturing meaningful patterns.

## 6.8 Purpose of Baseline Models

The baseline models are not expected to produce perfect predictions. Instead, they act as the first step in the modelling pipeline. They help reveal:

- How difficult the prediction task is,
- How much information simple features can provide and

- What performance level more advanced models should beat.

Once the baseline performance is measured, the next stage of the dissertation will focus on building more advanced models such as Random Forest and later interpreting feature importance and SHAP values.

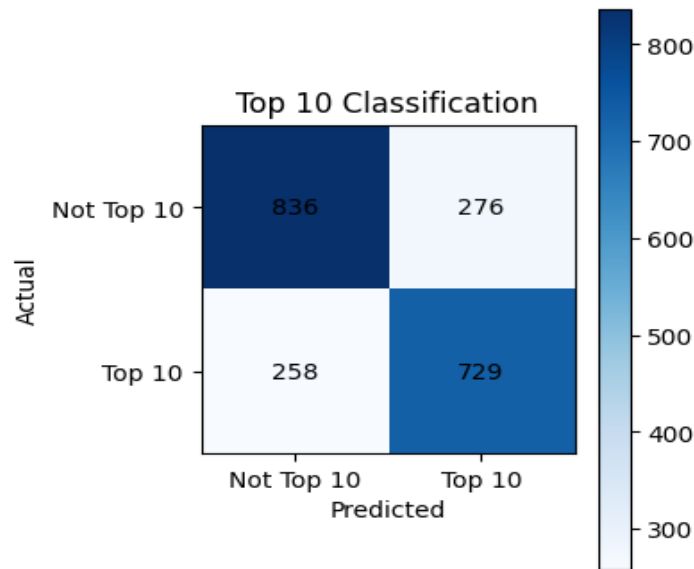
The models are designed to analyze and predict race outcomes using a combination of pre-race information and race execution variables, allowing the study to focus on performance modelling and factor importance rather than purely pre-race forecasting.

## 7 Results and Model Comparison

The first step in the modelling stage was to evaluate how well simple baseline algorithms could predict Formula 1 race outcomes. After that, I compared these results with more advanced models and examined which features had the biggest impact on predictions. Since the dissertation focuses on two tasks – predicting Top 10 finishes and predicting exact finishing positions – the results are also explained in that order.

### 7.1 Baseline Classification: Logistic Regression

The Logistic Regression model was used as the starting point for predicting whether a driver finishes inside the Top 10. The results showed an accuracy of around 0.74 and an F1-score of about 0.73. This means the model captures the main relationships in the data, especially the effect of qualifying and grid position. The confusion matrix shows that the model is balanced and does not heavily misclassify one of the classes. Even though it is a simple model, it already performs reasonably well, which indicates that the dataset contains strong signals about race outcomes.



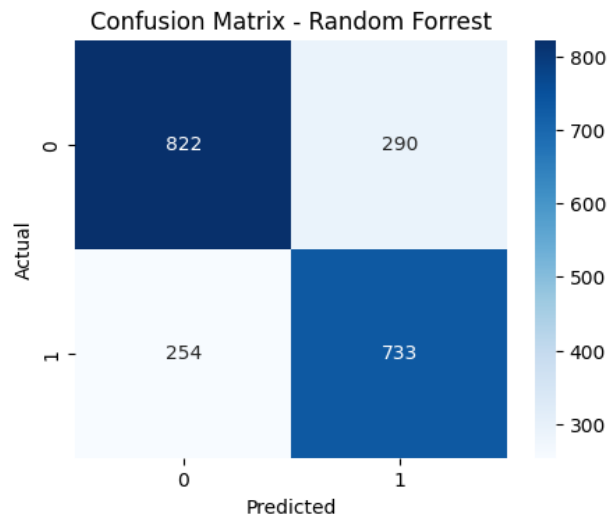
**Figure 7 Confusion Matrix of Logistic Regression**

## 7.2 Baseline Regression: Linear Regression

For predicting exact finishing positions, Linear Regression acted as the baseline. It achieved an RMSE of roughly 5.15, an MAE of about 4.13 and an R2-score of 0.33. These numbers mean that on average, the model is off by around four positions. Given how unpredictable races can be due to accidents, strategy changes and weather, this baseline is not surprising. Still, it helps set a clear benchmark for the more advanced models that follow.

## 7.3 Random Forest Classification

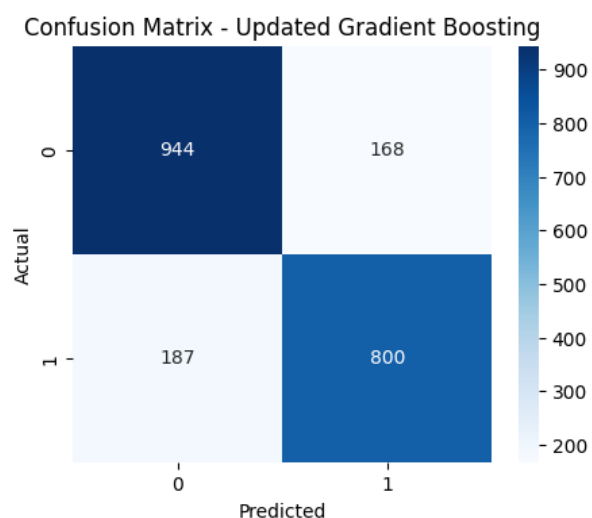
The first advanced model tested was the Random Forest (Staff, 2025b). It achieved an accuracy of about 0.74 and an F1-score of around 0.73. Surprisingly, these results were slightly worse than Logistic Regression. In theory, Random Forest should work better with mixed data and nonlinear patterns, but in this case the improvement was limited. One reason could be that the predictive relationships in the dataset are not extremely complex or that Random Forest needs more tuning to perform better. The confusion matrix also shows similar misclassification patterns to the baseline.



**Figure 8 Confusion Matrix of Random Forest Classification**

## 7.4 Gradient Boosting Classification

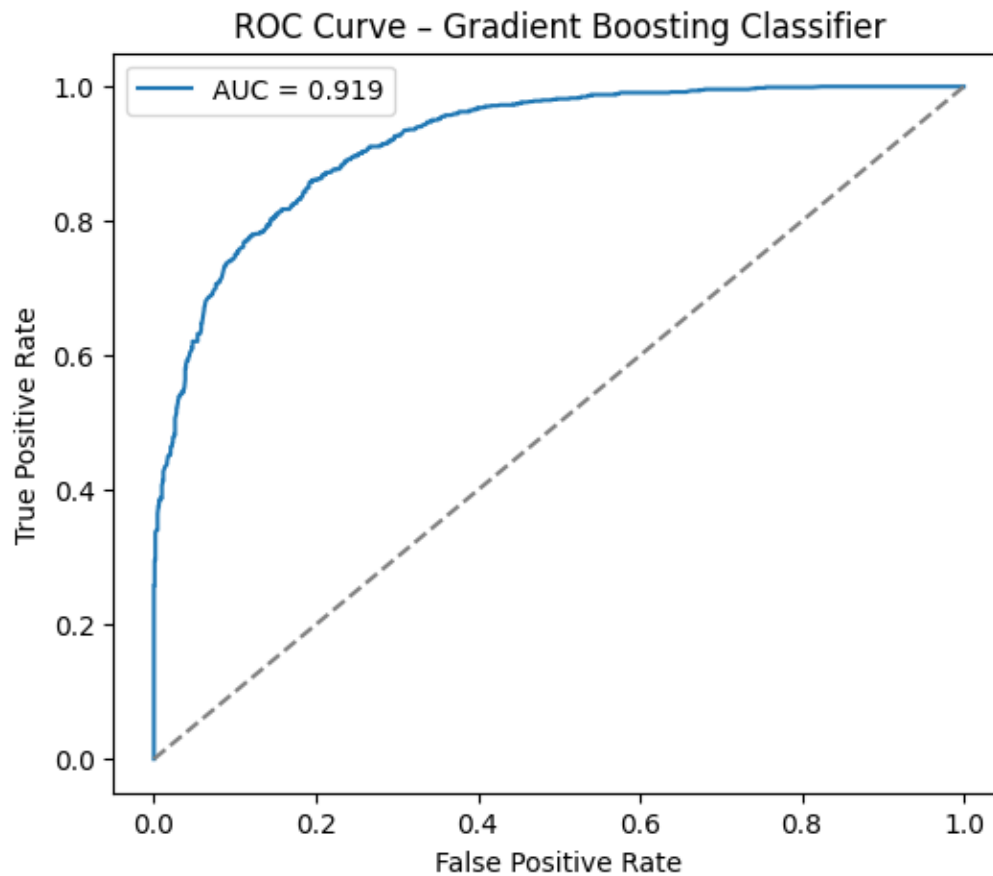
After the Random Forest, I trained a Gradient Boosting classifier (GeeksforGeeks, 2025b). With the original features, the model already performed better than the baseline and achieved an accuracy of around 0.76 and an F1-score of about 0.76. After adding the two engineered features, `driver_experience` and `recent_form_3`, the performance improved further. The final accuracy reached around 0.83 and the F1-score was around 0.82. This makes Gradient Boosting the best classification model in the project. The confusion matrix shows fewer incorrect predictions compared to the earlier models.



**Figure 9 Confusion Matrix of Gradient Boosting Classifier**

## 7.5 ROC Curve – Classification Model

Before moving to the regression results, I also checked how well Gradient Boosting classifier separates Top 10 and non-Top 10 finishers. The ROC curve clearly stays above the diagonal baseline, which means the model is doing a good job at distinguishing the two groups. The AUC score is around 0.92, so the classifier is strong across different probability thresholds, not only at the final prediction cut-off (Friedman, 2024). This confirms that the model is making reliable predictions and is not overly sensitive to one specific threshold.



**Figure 10 ROC Curve of Gradient Boosting Classifier**

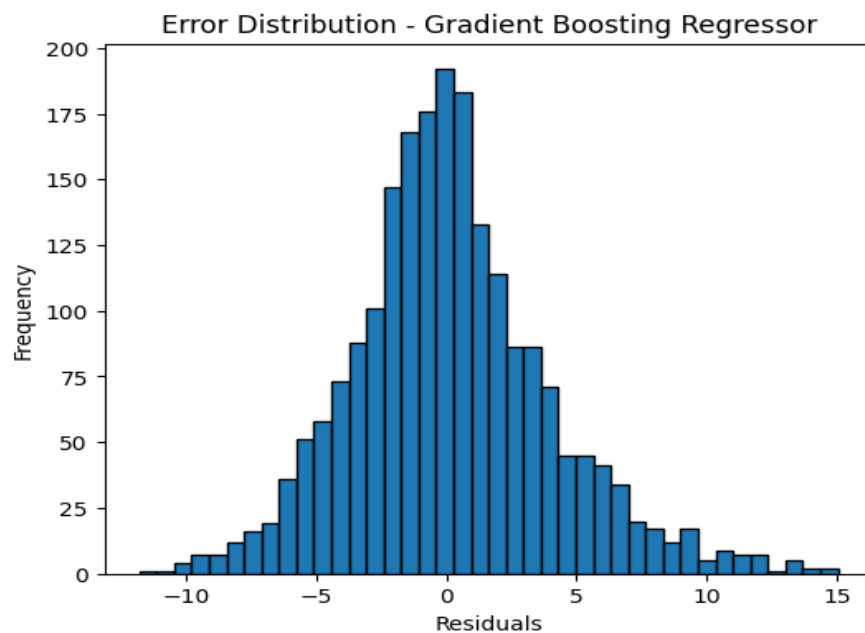
## 7.6 Gradient Boosting Regression

Gradient Boosting was also used for predicting exact finishing positions (GeeksforGeeks, 2025b). Before adding engineered features, the model reached an RMSE of about 4.89, an MAE of roughly 3.8 and an R2-score of around 0.4. These values were already better than the Linear Regression baseline. After adding the engineered features, the regression results improved further. The RMSE decreased to about 3.84, the MAE dropped to around 2.83 and

the R2-score increased to roughly 0.63. This means the model explains a much larger part of the variation in race results, which is significant given how unpredictable Formula 1 can be.

## 7.7 Error Distribution – Regression Model

I also looked at how the errors were distributed in the final Gradient Boosting regressor. Most of the residuals are close to zero, which means the model usually predicts finishing positions quite accurately. There are still some bigger errors on both sides, but this is normal in Formula 1 because unexpected events like crashes, safety cars or mechanical issues can change the race outcome immediately (Francis, 2025). Overall, the error distribution shows that the model captures the main behavior of race results and struggles mainly in unpredictable situations.



**Figure 11 Error Distribution of Gradient Boosting Regressor**

## 7.8 Why the Engineered Features Helped

Driver experience and recent form were added to capture information that does not appear directly in the original dataset. Driver experience helps describe how stable and consistent a driver becomes over time. Recent form shows how well a driver has been performing in the last few races. Formula 1 results often depend on confidence, consistency and short-term performance momentum, which can influence outcomes across races (Brolin, 2020; Hynes,



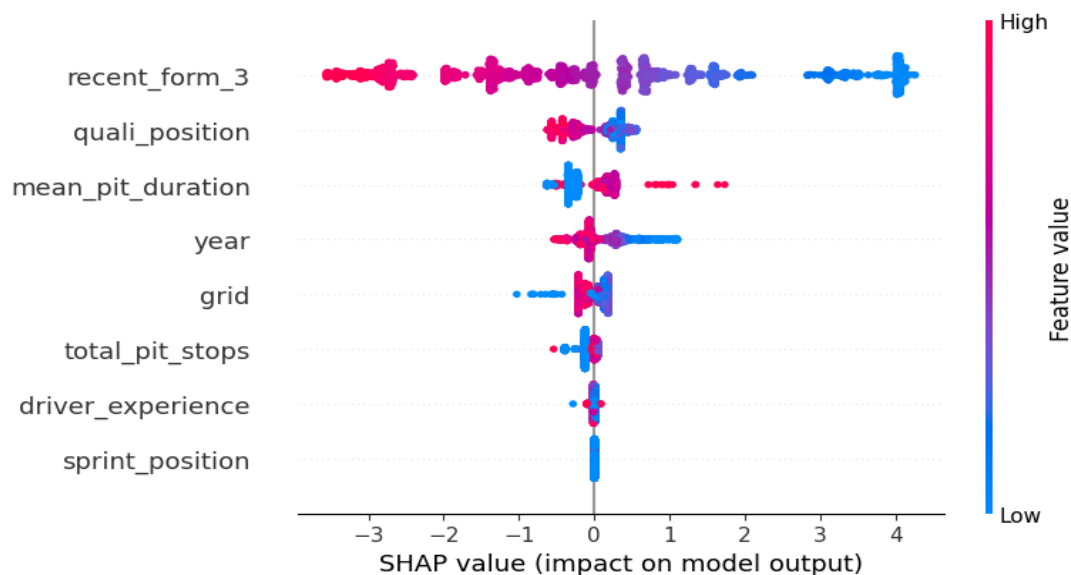
2024). This is also why both the classification and regression models improved after including them.

## 7.9 Feature Importance

Feature importance values from the Gradient Boosting models show which variables had the strongest influence on the predictions. In both models, qualifying position and grid position were the most important. This is expected, since starting near the front has a major impact on the final result. The engineered features also ranked high. Recent form helped identify drivers who were performing well at that stage of the season and experience added information about long-term performance. Total pit stops also appeared as an important feature in the regression model. More pit stops usually lead to worse results, which matches typical race strategy.

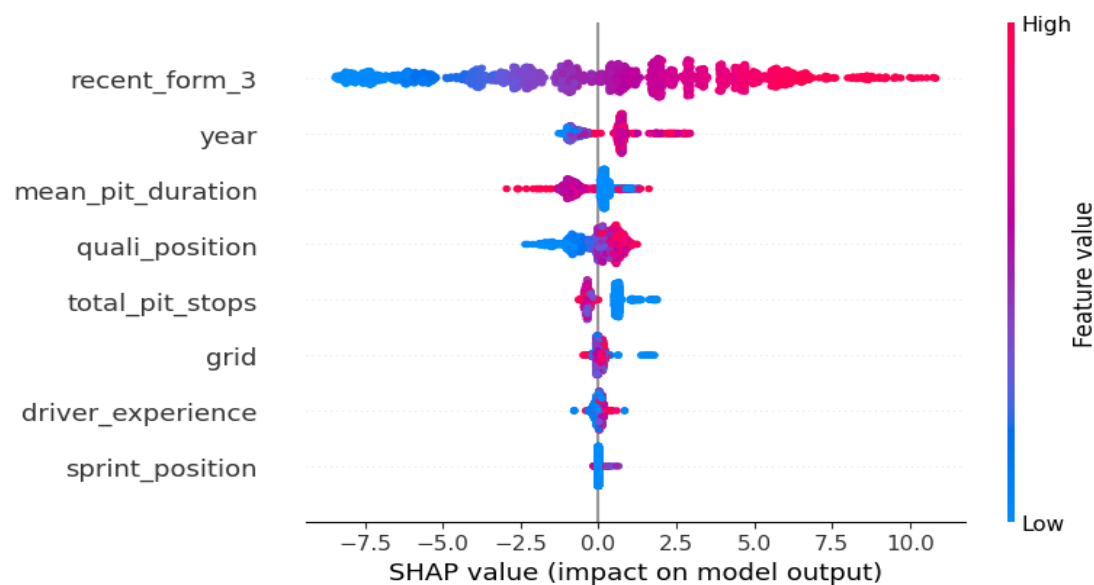
## 7.10 SHAP Interpretability

To understand the models more clearly, SHAP values were calculated. The summary plot for the classification model shows that `recent_form_3` has the strongest influence on the predictions. Qualifying position also has a clear effect. Mean pit stop duration and the year of the race contribute as well, while remaining features have smaller impacts.



**Figure 12 SHAP Values of Classification Model**

For the regression model, the SHAP plot shows that good recent form and strong qualifying results are linked to better predicted finishing positions. More pit stops and longer pit stops push the prediction toward a worse finishing position. Driver experience also supports better results. These patterns are consistent with normal Formula 1 performance.



**Figure 13 SHAP Values of Regression Model**

## 7.11 Summary of Model Performance

Overall, the modelling results show that more advanced algorithms were able to capture performance patterns in Formula 1 much more effectively than the baseline models. Logistic Regression provided a solid starting point for predicting Top 10 finishes and Linear Regression created a reasonable benchmark for predicting exact finishing positions. However, the Gradient Boosting models clearly performed the best in both tasks. After adding the engineered features, the classification model reached an accuracy of around 0.83 and the regression model achieved an R2 of roughly 0.63. These improvements show that features such as qualifying position, recent form and driver experience provide strong and consistent signals about race performance. Random Forest models, on the other hand, did not offer major improvements over baselines, which suggests that the relationships in the dataset are better captured by boosting methods. The comparison table below

summarizes the performance of each model and highlights how Gradient Boosting stands out as the most reliable approach for predicting race outcomes.

### 7.11.1 Model Comparison Table

| Top 10 Classification                                   |          |      |
|---------------------------------------------------------|----------|------|
| Model                                                   | Accuracy | F1   |
| Logistic Regression                                     | 0.74     | 0.73 |
| Random Forest Classifier                                | 0.74     | 0.73 |
| Gradient Boosting Classifier (with engineered features) | 0.83     | 0.82 |

| Finishing Position Regression                          |      |      |      |
|--------------------------------------------------------|------|------|------|
| Model                                                  | RMSE | MAE  | R2   |
| Linear Regression                                      | 5.15 | 4.13 | 0.33 |
| Random Forest Regressor                                | 5.19 | 3.95 | 0.32 |
| Gradient Boosting Regressor (with engineered features) | 3.84 | 2.89 | 0.63 |

## 8 Business Insights and Recommendations

The results of the modelling process do not only show which algorithms perform best, but they also reveal several practical insights that are relevant for teams, sponsors and decision-makers in Formula 1. While the models cannot predict every unexpected event, they highlight the key factors that consistently shape race outcomes. Based on the patterns observed in the data and the behaviour of the models, this chapter summarizes the most important business insights that can be applied in real Formula 1 contexts.

### 8.1 Qualifying Performance and Starting Position Remain the Strongest Predictors

Across all the models in this project, qualifying position appeared as one of the strongest predictors of race outcomes. The driver's starting position has a huge impact on how their

race will go. When a driver begins near the front, they avoid most of the traffic and problems that happen in the mid-field, which gives them a much better chance of scoring points.

Clark (2024) explains that the entire qualifying format is built to decide the starting grid as fairly as possible because starting position matters so much in modern F1. The knockout system (Q1-Q3) puts the fastest cars at the front, and this is important because overtaking is limited on many circuits. When overtaking is difficult, qualifying becomes one of the most decisive moments of the weekend.

Another important reason why qualifying matters is the “dirty air” effect. Dirty air refers to the turbulent airflow behind a car, which makes it much harder for the car behind to follow closely and overtake. Cook (2025) shows that dirty air reduces downforce, increases tire wear and destabilizes the car, especially during long stints behind another driver. Because of this, even a small difference in starting position can become a big advantage during the race.

The results of my models reflect these patterns very clearly. Drivers with better qualifying positions had a much higher chance of finishing inside Top 10 and the regression model predicted stronger finishing positions as well. This shows that qualifying performance is one of the most reliable indicators of race outcomes.

From a team perspective, this insight suggests a clear strategic direction: investing in single-lap performance – through aerodynamic upgrades, tire preparation and qualifying engine modes – should remain a priority because it directly increases the likelihood of scoring points. Additionally starting near the front reduces the effect of dirty air, limits the risk of collisions and gives teams a cleaner race environment.

From a sponsorship perspective, qualifying performance also affects visibility. Cars starting in the top positions appear more often on television during the early laps, which are usually the most watched moments of the race. Strong qualifying sessions therefore benefit teams and sponsors by increasing brand exposure.

## 8.2 Driver Momentum (Recent Form) Strongly Influences Race Results

One of the clearest findings from the engineered features was the importance of recent form. The `recent_form_3` variable – the driver’s average finishing position over the last three races – became one of the strongest predictors in both the classification and regression models. The SHAP results also confirmed that it had a greater impact than classic variables such as sprint position or pit stop counts.

This pattern reflects what happens in real Formula 1. Driver performance does not reset from one Grand Prix to the next. Confidence, mental state and adaptation to the car often create short-term performance cycles in which a driver either performs consistently well or repeatedly struggles. The Formula 1 official interview with Ricciardo and other drivers demonstrates this clearly. During his final months at McLaren, a loss of confidence led to what he described as a “mentally fatiguing spiral”, affecting his performance across multiple races. After taking a break and rebuilding his mindset, Ricciardo explained that returning with renewed confidence allowed him to perform more consistently again (Hynes, 2024).

The article also highlights how mental preparation, psychological support and stable focus help drivers maintain momentum across a long season. Sergio Perez, for example, used a sports psychologist to recover form after a run of poor results, highlighting how recent performance trends can influence future outcomes until the underlying issues are addressed. Performance coaches cited in the article similarly describe how drivers enter a “flow state” when confidence, focus and preparation align and this state often leads to a series of strong results rather than isolated performances.

For teams, this insight is important when evaluating mid-season strategy. If a driver enters a positive performance phase, it may be beneficial to prioritize upgrades or strategic opportunities toward that driver to maximize points. For sponsors, recent form is one of the most reliable indicators of upcoming visibility as strong results tend to persist over multiple races.

### 8.3 Driver Experience Supports More Consistent Performance

The second engineered feature – `driver_experience` – also improved the model and showed an important pattern. Drivers with more experience usually perform more steadily across a season. They seem to handle pressure better, avoid rushed decisions and adapt more easily when race conditions change. In both the classification and regression models, experience helped the predictions become more accurate.

This fits well with what happens in real Formula 1. At this level, a driver needs more than just raw speed. They must stay focused, manage stress and make good decisions in very intense situations, and these skills improve over years of racing. The Formula 1 article on mental preparation shows how experienced drivers like Kimi Raikkonen and Sebastian Vettel rely on strong routines and mental preparation techniques to stay calm and focused during qualifying and the race (Hynes, 2024). Performance coaches in the article explain that experienced drivers usually know exactly what they need to feel ready, which helps them stay consistent.

The article also highlights how important support networks are. Many experienced drivers work closely with long-term trainers; physios of team staff who help reduce distractions. This kind of stable environment makes it easier for them to stay focused and perform well each weekend. Drivers who are still learning the sport can sometimes struggle more with confidence or preparation, which can cause their results to change more from race to race.

For teams, this insight reinforces recommended approach: pairing a younger driver with an experienced teammate can stabilize race performance and improve development outcomes. For sponsors, experienced drivers offer more predictable performance and therefore more reliable visibility throughout the season.

## 8.4 Pit Stop Efficiency Still Matters for Race Outcomes

Although pit stops were not the strongest predictors in the machine learning models, they still showed a clear influence in results. The regression model and SHAP values both indicated that longer pit stops usually push the prediction toward a worse finishing position. This matches what happens in real Formula 1, where even a small delay in the pit lane can completely change the outcome of a race.

The article by Barretto (2019) explains that teams practice pit stops thousands of times because “race results depend on it”. Even a delay of half a second means something has gone wrong and this can be enough to lose several positions on track. Teams aim not only for very fast pit stops, but also for consistent ones, because consistency protects the driver’s race strategy.

The article also shows how much effort goes into achieving this. Pit crews train daily, both physically and technically, to reduce mistakes and stay sharp throughout the long season. Equipment such as jacks and wheel guns is constantly improved and even small details like pit-lane layout or the driver hitting their marks can make a difference. When everything works perfectly, pit stops like Red Bull’s famous 1.88-second stop are possible – but if anything goes wrong, the time loss directly affects the driver’s finishing position.

For teams, this insight highlights that investment in pit crew training and reliable equipment can translate into real points. For sponsors, efficient pit stops help maintain track position, which increases visibility and keeps the team in competitive positions throughout the race.

## 8.5 What This Means for Teams and Sponsors

The insights from the modelling stage can be directly applied to practical decision-making in Formula 1. Even though the models cannot predict every unpredictable event, they help explain which factors consistently matter for race performance. These patterns give teams and sponsors a clearer understanding of where competitive advantage really comes from and how race strategies can be optimized.

For teams, one of the most important messages is that performance trends do not happen by accident. The results showed that qualifying strength, recent form, experience and operational efficiency all contribute to predictable improvements in race outcomes. This means that teams can use these patterns to guide decisions during the season. For example, if a driver has a strong momentum, it may be beneficial to prioritize upgrades or strategic opportunities toward that driver, because the model shows that momentum is likely to continue. Similarly, improving qualifying performance – through car setup, aerodynamic development or tire preparation – can have a direct impact on scoring points and avoiding unnecessary risk in the midfield.

The models also highlighted the value of consistency. Drivers with higher experience and strong recent performance tended to achieve better predicted outcomes and this makes them important assets for team stability. Many constructors already choose to pair a younger driver with a highly experienced one and the model results support that approach. Stable performance reduces strategic uncertainty, improves car development feedback and helps teams score points more reliably over a long season.

From a sponsorship and commercial perspective, these results also carry clear implications. Strong qualifying performance increases early-race visibility, which is when broadcast viewership is usually highest. Drivers in good recent form also gain more attention because positive results attract media coverage across several races, not just one. Experienced drivers, meanwhile, offer predictable exposure and lower risk, which is valuable for long-term partnerships.

Overall, the modelling results suggest that performance in Formula 1 is shaped by a combination of car pace, driver momentum, experience and operational execution. Teams can use these insights to refine their strategies and sponsors can use them to make more confident decisions about partnerships and visibility. Even though Formula 1 will always involve unpredictability, understanding these patterns helps everyone involved make more informed and effective decisions.

## 9 Conclusion and Limitations

This dissertation explored how data analytics and machine learning can be used to understand and predict performance in Formula 1. The main goal was to analyze historical race data, build predictive models for both Top 10 finishes and exact race positions and explain which factors matter most in shaping results. Alongside this technical work, the project also connected these findings to the business side of Formula 1, showing how performance patterns translate into sponsorship value, strategic planning and competitive advantage.

The dataset from Kaggle provided a large historical base, covering more than seventy years of racing. After combining multiple tables and engineering new features, a clean race-level dataset was created where each row represented one driver in one race. Exploratory analysis confirmed several well-known patterns in Formula 1: qualifying position has a strong impact on finishing results, strategy and pit stops play a meaningful role and long-term performance cycles reflects regulatory changes and team development. These insights helped shape expectations for the modelling stage.

Baseline models established a clear reference point. Logistic Regression performed reasonably well for Top 10 prediction and Linear Regression captured a meaningful part of the finishing position patterns, even though race results are highly variable. The more advanced Gradient Boosting models, especially after adding the engineered features (driver\_experience and recent\_form\_3), demonstrated significantly stronger predictive performance. The final classification model reached around 0.83 accuracy, while regression model achieved an R2 value of roughly 0.63. These results show that machine learning can identify and model many of the key factors that shape Formula 1 outcomes.

Model interpretation tools, such as feature importance and SHAP, helped connect the technical results with practical explanations. Qualifying position, recent performance and driver experience consistently appeared as the strongest predictors of race success. Pit stop efficiency also had a noticeable impact, especially in the regression model. These findings match real-world behavior in Formula 1, where grid position, momentum and execution play critical roles across a long season.

Although the models performed well, there are natural limitations. Race results depend on many unpredictable and situational factors – weather changes, mechanical failures, safety-car timing, accidents and mid-season car upgrades (Francis, 2025) – that were not included in the dataset. These elements can change the outcome of a race instantly and are difficult to capture using historical data alone. Another limitation is the wide time span of the dataset: Formula 1 has changed dramatically across decades, with new regulations, race



formats and scoring systems. Even though the models generalized well, performance patterns from the 1970s or 1980s cannot fully match the structure of modern Formula 1. This means that some variability in the predictions comes from differences between eras rather than modelling errors.

Despite these constraints, the project demonstrates that a large portion of race performance can be modelled and explained using historical data. With additional information – such as weather, tire compounds, team upgrade cycles or lap-by-lap pace – predictive accuracy could be improved further. However, even in its current form, the analysis provides both technical understanding and business-relevant insights, showing how competitive success in Formula 1 reflects a combination of engineering strength, driver momentum, strategy and operational execution.

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## Appendix

Appendix A: GitHub Repository:

The complete codebase used for data preprocessing, exploratory analysis, feature engineering, model training, evaluation and SHAP analysis is available in the following GitHub repository:

GitHub repository: <https://github.com/giorgi00798/Business-Data-Analytics-Project>